CSC 6740 **Data mining** **D**ue by: Oct. 23th

Name:

Q1: Using the data in the table to build a decision tree classifier **step by step**, with information gain (if your panther ID is even), GINI index (if your panther ID is odd) to classifier C0/ C1 classes. The max depth is 2. **(20 points)**

Customer ID	Gender	Car Type	Shirt Size	Class
1	M	Family	Small	C0
2	M	Sports	Medium	C0
3	M	Sports	Medium	C0
4	M	Sports	Large	C0
5	M	Sports	Extra Large	C0
6	M	Sports	Extra Large	C0
7	F	Sports	Small	C0
8	F	Sports	Small	C0
9	F	Sports	Medium	C0
10	F	Luxury	Large	C0
11	M	Family	Large	C1
12	M	Family	Extra Large	C1
13	M	Family	Medium	C1
14	M	Luxury	Extra Large	C1
15	F	Luxury	Small	C1
16	F	Luxury	Small	C1
17	F	Luxury	Medium	C1
18	F	Luxury	Medium	C1
19	F	Luxury	Medium	C1
20	F	Luxury	Large	C1

Q2: How to overcome the overfitting issue in the decision tree classifier? please give more details how will you select the parameters used for each strategy in practice. **(10 points)**

Q3: Use naïve bayes classifier to predict the new data positive/negative class with consideration avoiding zero probability condition by Laplace estimate. New data: **[A=1, B=0,C=0] which class? Show step by step procedure. (20 points)**

Record	A	B	C	Class
1	0	0	0	+
2	0	0	1	-
3	0	1	1	-
4	0	1	1	-
5	0	0	1	+
6	1	0	1	+
7	1	0	1	-
8	1	0	1	-
9	1	1	1	+
10	1	0	1	+

CSC 6740 **Data mining** **D**ue by: Oct. 23th

Name:

Q4: A manufacturing company wants to automatically classify finished units as **good** or **bad** quality. Two analysts have each built a classifier from the same historical inspection data of **1,000 records** (850 good, 150 bad):

- Analyst A built a **decision tree** (C4.5-style, with pessimistic-error pruning).
- Analyst B built a **naive Bayes classifier** (with Laplace smoothing).

Your task is to design a fair comparison of the two classifiers and determine, using the results below, whether one performs significantly better than the other. Treat **bad** as the positive class throughout.

Table 1. Error rates (%) from stratified 10-fold cross-validation. Both classifiers used the same folds; each test fold has 100 records (85 good, 15 bad).

Fold	1	2	3	4	5	6	7	8	9	10
Decision tree	12	14	11	15	13	12	16	13	12	14
Naive Bayes	14	15	14	14	15	12	19	14	14	15

Table 2. Confusion matrices pooled over all 10 test folds (n = 1,000).

	DT: predicted bad	DT: predicted good		NB: predicted bad	NB: predicted good
Actually bad	105	45	Actually bad	126	24
Actually good	87	763	Actually good	122	728

1. Analyst A reports that the decision tree reached 98% accuracy on the data it was trained on. Explain why this number can't be used to compare the classifiers. **(5 points)**
2. Explain why the cross-validation should be **stratified** for this dataset, and why both classifiers must be evaluated on **the same folds for paired-t-test**. **(5 points)**
3. Both analysts tuned hyperparameters (the pruning confidence factor and the smoothing strength). Describe a procedure that tunes them without biasing the error estimates, and name the problem your procedure prevents. (**5 points)**

4. From Table 2, compute for each classifier: accuracy, error rate, precision, recall (sensitivity), specificity and F1 for the bad class. Present the results in a table. **(5 points)**

5. Using Table 1, carry out a **paired t-test** at $\alpha = 0.05$ on the per-fold differences in error rate. State H_0 and H_1, compute t statistic, give the degrees of freedom and the critical value (attached at the end of assignment), and state your conclusion.**(10 points)**

CSC 6740 **Data mining** **D**ue by: Oct. 23th

Name:

Q5: The data tuples of Table right are sorted by decreasing probability value, as returned by a classifier to be classified as positive. We test a list of threshold used to make final positive /negative calls. Threshold list is [0.4, 0.50, 0.55, 0.6, 0.65, 0.7, 0.8, 0.9] , data tuples with Prob.>=Threshold will be **P**ositive, otherwise **N**egative.

Find TP, FP, TN, FN, and Sensitivity (TP Rate: TPR), Specificity (TN Rate: TNR), and False Positive(FPR: 1-TNR), for each threshold, fill in the table below. Draw the ROC curve and compute AUC. **(10 points)**

Tuple #	*Class (True)*	*Prob.(classifier)*
1	p	0.95
2	n	0.85
3	p	0.78
4	p	0.66
5	n	0.60
6	p	0.55
7	n	0.53
8	n	0.52
9	n	0.51
10	p	0.4

Threshold	TP	FP	TN	FN	TPR	TNR	FPR
0.90							
0.80							
0.70							
0.65							
0.60							
0.55							
0.50							
0.40							

Q6: Descibe the Adaboost algorithm using decision tree as a base classifier **(10 points)**

Step1 :

Step2:

CSC 6740 **Data mining** **D**ue by: Oct. 23th

Name:

t Table

cum. prob	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$	$t_{.995}$	$t_{.999}$	$t_{.9995}$
one-tail	0.50	0.25	0.20	0.15	0.10	0.05	0.025	0.01	0.005	0.001	0.0005
two-tails	1.00	0.50	0.40	0.30	0.20	0.10	0.05	0.02	0.01	0.002	0.001
df											
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.000	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.000	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.000	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.000	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.000	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.000	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.000	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.000	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.000	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.000	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	0.000	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	0.000	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
80	0.000	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
100	0.000	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390
1000	0.000	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300
z	0.000	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291
	0%	50%	60%	70%	80%	90%	95%	98%	99%	99.8%	99.9%
	Confidence Level										